

Appendix

All but one of the tables that appear in chapters 4, 5, 6, and 8 summarize parts of a content analysis of empirical articles published in psychological journals during the discipline's early years. The selection of journal volumes and articles for inclusion in this analysis was based on the following considerations.

Four basic reference points – 1895, 1910, 1925, and 1935 – were chosen for the purpose of establishing time trends. Journal volumes published during those years were included in the analysis as were the immediately preceding and the immediately following volume. For each set of three consecutive volumes of a journal, the results of the analysis were pooled to allow for minor fluctuations from one year to the next. This yielded data for the periods 1894–1896, 1909–1911, 1924–1926, and 1934–1936. There were minor discrepancies, as when the results of the analysis of the first three volumes of a journal that began publication in 1910 were included in the 1909–1911 period.

Only one of the journals analyzed, the *American Journal of Psychology*, provided a source of empirical articles over the entire period. The other journals began publication at a later stage or ceased publishing empirical studies in significant numbers at some point. The second decade of the twentieth century was a particularly fertile one in terms of the appearance of new American journals. For tracing changes in these journals the 1910 reference point was therefore impractical. Interruption of publication around 1918 introduced a further complication in some cases. To deal with these problems two procedures were adopted. For journals starting publication in this decade the first three volumes were the ones analyzed, irrespective of year of publication. For older journals the analysis of the 1909–1911 period was supplemented by an analysis of volumes published between 1914 and 1916 to reduce the time gap between volumes of these

journals and volumes of journals that began publication later in the decade. These data are referred to as the 1914–1916 period in tables 5.3 and 5.4, and the 1914–1920 period in table 8.1.

The following categories of journal article were not included in the analysis: theoretical articles; reviews of previously published work; articles devoted entirely to the description of apparatus; brief notes, generally not more than two pages in length; and articles based on work using only nonhuman subjects. Agreement among raters about articles to be excluded was 97 percent.

In most cases all eligible articles in a journal volume were coded. However, a random sampling of eligible articles was used in the case of the 1949–1951 time period appearing in table 5.3 and also in the case of the two applied journals for the 1924–1926 and 1934–1936 periods where coding categories were those discussed in chapter 6.

The percentages in table 6.1 are expressed in terms of the number of empirical studies using human subjects. This includes studies containing inadequate or no background information on subjects, which account for about 22 percent of all studies during the earlier periods and about 13 percent during the later periods. This would tend to depress row totals below 100. However, some studies employed more than one category of subject, thus raising row totals. The two effects tend to cancel each other out. Subjects identified as graduate students are not included in this tabulation. The category “noneducational” refers to adult subjects drawn from anywhere but an educational institution. The summary data presented here are based on an analysis of three volumes of each of six journals for the later periods, fewer for the earlier periods when some journals had not yet appeared. The journals are those identified in tables 5.2, 5.3, and 5.4.

Notes

Chapter 1. Introduction

1 To a considerable extent this is also true of physical science; see P. Janich, "Physics – natural science or technology?" in W. Krohn, E. T. Layton, Jr., and P. Weingart, eds., *The Dynamics of Science and Technology* (Dordrecht: Reidel, 1978), pp. 3–27.

2 For a general critique of the distinction between the two contexts, see P. Feyrabend, *Against Method* (London: NLB, 1975), pp. 165–169; and T. Nickles, "Scientific discovery and the future of philosophy of science," in T. Nickles, ed., *Scientific Discovery, Logic and Rationality*, Boston Studies in the Philosophy of Science, vol. 56 (Dordrecht and Boston: Reidel, 1980), pp. 1–59.

3 See S. Bem, "Context of discovery and conceptual history of psychology," in S. Bem, H. Rappard, and W. van Hoorn, eds., *Proceedings of the First European Meeting of Cheiron* (Leiden: Ryksuniversiteit, 1983), pp. 207–231.

4 "Discoveries do not simply 'occur' or 'happen' naturalistically, but are socially defined and recognized productions. In other words, the question is not what makes them happen, but rather what makes them discoveries." See A. Brannigan, *The Social Basis of Scientific Discoveries* (Cambridge: Cambridge University Press, 1981), p. 77.

5 cf. K. J. Gergen, *Toward Transformation in Social Knowledge* (New York: Springer, 1982), p. 101: "The audience for research reports is never exposed to ongoing events; one never gains first-hand experience with the research process itself. Rather, the chief product of research is language. Research reports essentially furnish linguistic accounts or interpretations."

6 A useful analysis of such constructive schemes, with illustrations from the history of experimental psychology, has been presented by Gernot Böhme, "The social function of cognitive structures: A concept of the scientific community within a theory of action," in K. D. Knorr, H. Strasser, and H. G. Zillian, eds., *Determinants and Controls of Scientific Development* (Dordrecht: Reidel, 1975), pp. 205–225; and G. Böhme, "Cognitive norms, knowledge interests and the constitution of the scientific object: A case study in the functioning of rules for experimentation," in E. Mendelsohn, P. Weingart, and R. Whitley, eds., *The Social Production of Scientific Knowledge* (Dordrecht: Reidel, 1977), pp. 129–141.